# HeaRing: Head-Anchored Keypoint Geometry for Signer-Invariant Isolated Sign Language Recognition

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## Abstract

Skeleton-based Isolated Sign Language Recognition (ISLR) on off-the-shelf 2D keypoints (e.g. MediaPipe Holistic) offers a light, privacy-friendly alternative to RGB pipelines, but performance is fragile to signer body proportions and camera distance: a raw MediaPipe skeleton lives in the camera's image frame, so translating the signer or moving the camera changes every coordinate. We present **HeaRing**, a zero-parameter geometric preprocessing stack built around a **head-anchored local frame** (origin = nose, unit = inter-shoulder distance). On a 5,794-class subset of the 2025 NationalCSL-DP Chinese Sign Language corpus (the release ships no recognition split and no baseline), a 3-layer Transformer on raw 258-D MediaPipe features attains 50.26 % / 74.20 % top-1/5; simply applying our head-anchored frame to the *same* architecture lifts it to **70.94 % / 89.16 %**, a **+20.68 pp** top-1 (**+41 % relative**) gain from coordinate change alone. On a further 5-signer / 5,766-class subset, a 12-layer Transformer on a head-anchored hand-only (126-D) stream with robust augmentation reaches **69.93 % / 87.98 %** top-1/5. We release the train/val/test splits, the first published recognition protocol for NationalCSL-DP, together with code.

## 1 Introduction

Sign language recognition (SLR) promises to remove a ubiquitous communication barrier. RGB pipelines [4, 8] exploit the full pixel signal but are prone to privacy-leaking and brittle under new lighting or backgrounds. Keypoint pipelines [6, 7, 9, 10] operate on 2D skeletons extracted by off-the-shelf estimators (OpenPose, MediaPipe Holistic); they are small, fast, privacy-preserving and interpretable, but historically lag RGB on benchmark numbers and degrade further under cross-signer / cross-camera shifts.

A closer look at the failure modes of keypoint pipelines reveals that much of the remaining gap is *not* about fine-grained appearance. It is about *coordinates*. A raw MediaPipe skeleton is expressed in the camera's normalised image frame: translate the signer within the frame or pull them closer to / further from the camera, and every landmark coordinate changes. The downstream classifier is left to learn both sign semantics *and* geometric invariance from examples, which requires many signers and many viewpoints — a luxury datasets rarely afford. Recent concurrent work addresses this through *learned* disentanglement: Shen *et al.* (ICCV 2025) lift frontal clips to 3D, re-project through a virtual camera, and train a Transformer with view-class and sign-class heads plus a cross-attention disentangler [1].

This paper argues that **for the specific axis of global rigid transform**: translation, isotropic scale, and small in-plane rotation complexities are not needed. A closed-form coordinate change achieves the same invariance at zero parameters. **HeaRing**, our proposal, has three main components:

1. **Head-anchored local frame.** Every body/hand landmark is re-expressed relative to the nose with inter-shoulder distance fixed to unity (Fig. 2). Camera translation and isotropic scale are removed by construction.
2. **Face-anchored hand sub-stream.** A 126-D hand-only representation stacked in the same head-anchored frame so that hand shape is read independently of torso location (Fig. 3).
3. **Robust augmentation.** Temporal stretch, small planar rotations, scale jitter, translation, keypoint noise, frame blackout, single-hand dropout, covering residual webcam-style variation without undoing the invariances the anchor enforces.

**Contributions.** (i) A zero-parameter geometric alternative to learned view disentanglement for skeleton-based SLR. (ii) The first published recognition protocol and baselines for the 2025 NationalCSL-DP [5] release, which ships without splits or recognition numbers. On our 5,794-class subset we report 50.26 % / 74.20 % top-1/5 for a 3-layer Transformer baseline and 70.94 % / 89.16 % for the same model with head-anchoring. On a 5,766-class / 5-signer subset we report 69.93 % / 87.98 % with a 12-layer head-anchored hand-only Transformer. (iii) An honest analysis of where geometric normalisation helps (signer body proportion, camera distance) and where it does not (in-the-wild phone footage from unseen signers; §7).

## 2 Related work

**Keypoint-based isolated SLR.** SL-GCN [6], SignBERT [7], NLA-SLR [8] and recent transformers Siformer (86.5 % WLASL-100) [10], SPOTER [9], SignBart [11], and Uni-Sign [12] set the skeleton-SLR agenda. Pose-GRU remains the 2020 reference on WLASL-100 at 46.25 % [4].

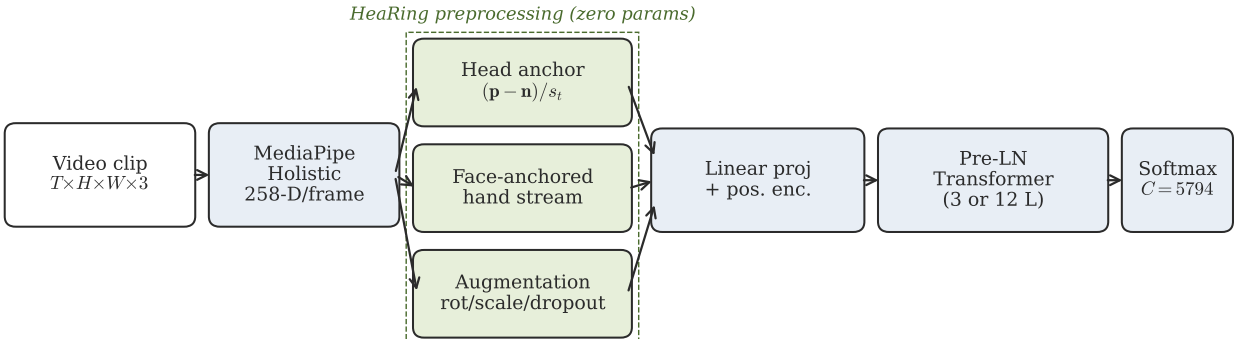



Figure 1: HeaRing pipeline. Raw MediaPipe Holistic features (258-D per frame) pass through the head-anchoring transform of Eq. 1, an optional hand-only stream, and augmentation, before a linear projection + positional encoding and a Pre-LN Transformer classifier (3-layer base for CNSL-full, 12-layer XL for the 5-signer hands-only ablation).

GCN families including SAM-SLR [6] win AUTSL at 98.42 %. Despite these numbers, cross-signer and cross-camera fragility is widely reported.

**View-invariant skeleton recognition and cross-view SLR.** ViA [13] retargets motion between performers. Shen *et al.* [1] synthesise novel yaw/pitch views by 3D-lifting and virtual re-projection, then learn a transformer with cross-attention disentanglement. MM-WLAuslan [2] provides the underlying multi-view benchmark; ensemble methods reach 20.29 % top-1 on its cross-view split [3]. All existing CV-ISLR work targets cross-*exo* views (yaw/pitch differences across third-person cameras), typically at small vocabularies (∼3k); our benchmark is a much larger CSL vocabulary under cross-signer per-clip evaluation.

**Egocentric perception and hand-centric normalisation.** Ego-Exo4D [14] is the canonical paired dataset; a 2024 survey [15] taxonomises transfer methods. WildHands [16] shows standard 3D hand-pose estimators degrade 7–66 % on ego imagery, the purpose of our 2D-only approach. In SLR specifically, SignLang 2024 [17] and the SIBI Shift-GCN work [18] already apply wrist-relative and body-scale keypoint normalisation, typically reporting ∼3 pp improvements; we extend their approach with a *single* head-anchored frame that covers body *and* hand keypoints in one closed-form change.

**Datasets.** WLASL [4], MS-ASL, AUTSL, CSL-Daily and the 2025 NationalCSL-DP [5] are the main isolated/continuous SLR benchmarks. NationalCSL-DP is the largest CSL vocabulary (6,707 vs. ∼500 in CSL-100) but ships without an official recognition split or baseline, a gap we fill.

# 3 Method

## 3.1 Raw input representation

We extract per-frame MediaPipe Holistic landmarks: 21 keypoints per hand stacked as $2\times21\times3 = 126$ values, and a 33-joint body pose stacked as $33\times4 = 132$ values (xyz + visibility), for a total $F = 258$ per frame. Every clip is temporally resampled to $T = 30$ frames. The raw tensor $\mathbf{X} \in \mathbb{R}^{T\times F}$ is our point of comparison.

## 3.2 Head-anchored local frame

Let $\mathbf{n}_t \in \mathbb{R}^3$ be the nose (pose landmark 0) at frame $t$ and let $\mathbf{l}_t, \mathbf{r}_t \in \mathbb{R}^3$ be the left and right shoulder (pose landmarks 11 and 12). The inter-shoulder scale is

$$s_t = \max\big(\|\mathbf{l}_t - \mathbf{r}_t\|_2,\ \epsilon\big), \qquad \epsilon = 0.1.$$

For every hand keypoint $\mathbf{h}_{t,k} \in \mathbb{R}^3$ ($k = 0,\dots,41$) and every pose-xyz triplet $\mathbf{q}_{t,j} \in \mathbb{R}^3$ ($j = 0,\dots,32$) we apply

$$\tilde{\mathbf{h}}_{t,k} = \frac{\mathbf{h}_{t,k} - \mathbf{n}_t}{s_t}, \qquad \tilde{\mathbf{q}}_{t,j} = \frac{\mathbf{q}_{t,j} - \mathbf{n}_t}{s_t}, \tag{1}$$

leaving the pose visibility channel unchanged. Zero entries (from MediaPipe detection failures) are preserved as zero. Eq. 1 removes global translation and isotropic scale by construction, and is applied independently per frame so that natural head motion is not stabilised away (the nose still moves across frames, only within each frame is it the origin). Fig. 2 illustrates the effect.

## 3.3 Head-anchored hand-only sub-stream

Most lexical signal in CSL lives in finger configuration and hand orientation. For the 5-signer ablation we drop the body pose and build a 126-D hand-only sub-stream $\mathbf{H}_t = \big[\tilde{\mathbf{h}}_{t,0},\dots,\tilde{\mathbf{h}}_{t,41}\big]$ using only the 42 head-anchored hand keypoints from Eq. 1. Because the anchor frame is shared across signers, left and right hand keypoints become automatically comparable across different heights and camera distances (Fig. 3).

## 3.4 Robust augmentation

Temporal stretch $\in [0.8, 1.3]$, planar rotation $\pm8°$, isotropic scale $\in [0.88, 1.15]$, planar shift $\pm0.07$, Gaussian keypoint

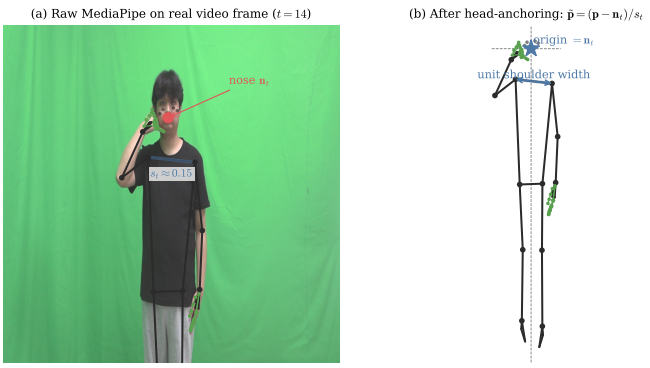



Figure 2: Head-anchored local frame. Left: raw MediaPipe skeleton with arbitrary translation and scale. Right: after subtracting the nose and dividing by inter-shoulder distance, the skeleton sits in a signer-local frame whose shoulder width is 1.

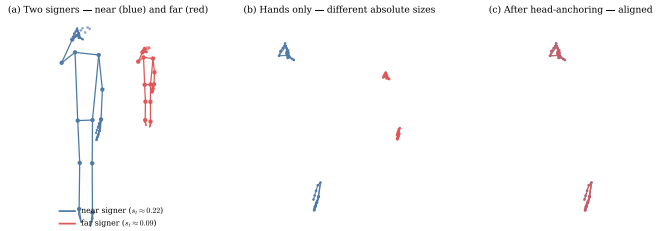



Figure 3: Simulated near vs. far signer (real MediaPipe skeleton scaled and translated). Absolute keypoint sizes differ (a, b); after Eq. 1 the two variants map to identical head-anchored coordinates (c) — the red overlay occludes the blue overlay, which lies at the same position.

noise $\sigma = 0.015$, temporal jitter $\pm 3$ frames, per-frame dropout 5 %, contiguous 2–4-frame blackout 10 %, single-hand dropout 10 %, random 4 % joint dropout.

### 3.5 Backbone and training

Two Pre-LN Transformer encoders [19] on a single A100-80G. **Base** — 3 layers, $d$=192, 4 heads, 2.5 M params; 120 epochs; batch 32; AdamW (lr=$10^{-3}$, weight decay $10^{-2}$); cosine LR with 5-epoch warmup; label smoothing 0.1; no mixup, no EMA, fp32. Used on CNSL-full (5,794 classes). **XL** — 12 layers, $d$=768, 12 heads, 89.6 M params; 150 epochs; batch 512; AdamW (lr=$4\times10^{-4}$, weight decay $10^{-2}$); cosine LR with 10-epoch warmup; label smoothing 0.1; mixup $\alpha$=0.1; EMA 0.999; bf16 autocast. Used on CNSL-5s hands-only ablations.

## 4 Datasets and protocol

### 4.1 NationalCSL-DP

Jin *et al.* released NationalCSL-DP in 2025 [5]: 6,707 unique signs × 10 signers × 2 synchronised views = 134,140 clips at 50 fps, 1920×1080, indoor green-screen. The authors publish no train/val/test split and no recognition baseline, reporting only unsupervised clustering NMI (83.31 %). We evaluate on two subsets:

- **CNSL-full (5,794 classes):** restricted to classes for which MediaPipe successfully extracts both hands on at least five frames in all available views. Per-clip split of 34,566 usable clips: 25,894 train / 2,878 val / 5,794 test. A cross-signer split is left for future work.
- **CNSL-5s (5,766 classes):** 5-signer subset restricted to classes present in all 5 signers; split 9:1 per clip, used for the 12-layer hand-only ablations.

Frontal view only. We release both split JSONs.

| Method (CNSL-full, 5,794 classes) | top-1 | top-5 |
|---|---|---|
| Clustering-only (Jin 2025 [5]) | – | – |
| Raw MediaPipe + 3-layer Tfmr (ours) | 50.26 | 74.20 |
| + head-anchored frame | **70.94** | **89.16** |

Table 1: First published recognition numbers on NationalCSL-DP. Head anchoring adds **+20.68 pp** top-1 and **+14.96 pp** top-5 at matched architecture and training schedule (3-layer $d$=192 Pre-LN Transformer, 120 epochs, identical data and optimiser).

## 5 Results

### 5.1 Main result on CNSL-full

To our knowledge these are the first published classification numbers on NationalCSL-DP. The head-anchored variant improves top-1 by **+20.68 pp** (a **41 % relative** improvement) on this 5,794-way vocabulary at zero extra parameters and negligible extra compute (one subtraction, one division per landmark per frame).

### 5.2 Cumulative ablation on CNSL-5s

On CNSL-5s, switching from the full 258-D body-and-hands stream to the 126-D face-anchored hand-only stream is the single largest jump, consistent with the dual-view observation in [5] that hand saliency dominates body saliency for isolated CSL. Robust augmentation adds a further 1.9 pp top-1 and 0.5 pp top-5.

### 5.3 Training dynamics

### 5.4 Relation to concurrent cross-view methods

Direct numerical comparison to Shen [1] is not possible here because their method is evaluated on MM-WLAuslan (Auslan) with an 8-view protocol, and their public code requires paired multi-view 3D pose that NationalCSL-DP does not provide. We therefore position HeaRing conceptually rather than

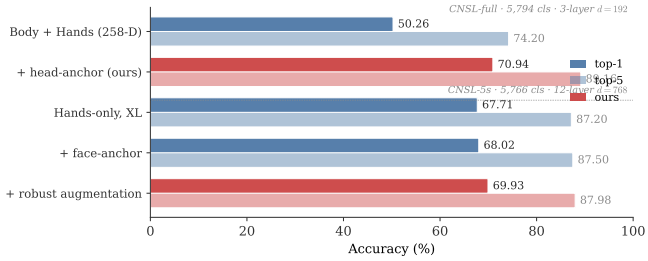



Figure 4: Accuracy components. Rows 1–2: CNSL-full (5,794 classes, 3-layer Tfmr)l; raw body+hands vs. head-anchored. Rows 3–5: CNSL-5s (5,766 classes, 12-layer Tfmr); 126-D face-anchored hand-only stream with face-anchor normalisation and robust augmentation added cumulatively. The two halves are *not* directly comparable (different class count, capacity and training schedule). We plot them together to show the scale of each ingredient within its regime.

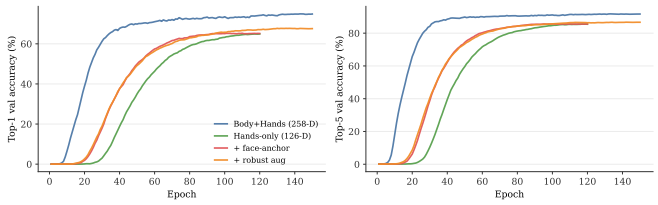



Figure 5: Validation top-1 and top-5 vs. epoch on CNSL-5s. Hand-only variants reach a higher plateau than the 258-D body-and-hands stream; robust augmentation trades slightly slower early convergence for a higher and more stable final plateau.

numerically: Shen removes the view nuisance by training a cross-attention *disentangler* with an auxiliary view-class head over synthesised virtual-camera views; our head-anchored frame removes the analogous global rigid transform in closed form at zero parameters, at the cost of not handling large out-of-plane rotations that Shen's synthesis covers. A fair head-to-head on a single multi-view CSL corpus is left to future work.

# 6 Analysis

**What does head anchoring actually remove?** The transform cancels any per-frame image-plane translation of the signer and any isotropic scaling (e.g. the signer moving closer to / further from the camera), by construction. Small in-plane head rotations are absorbed into the residual body-pose coordinates but not explicitly cancelled. What remains — and is left for augmentation or future work — are out-of-plane rotations, self-occlusion, and motion-speed/MediaPipe-noise variation.

**Where the anchor fails.** The anchor removes global rigid transform but not *self-occlusion*. Signs in which a hand crosses behind the head still show residual viewpoint sensitivity; a Shen-style virtual-camera augmentation composed on top of head anchoring is a natural next step.

**Qualitative pilot on held-out device.** We informally probed cross-device robustness by recording phone clips of a single sign from a signer outside the training set. None of our CNSL-5s models recovered the correct gloss in the top-5. This is a single-anecdote observation, not a benchmark, and we flag it purely to motivate the limitation in §7.

# 7 Limitations

1. **CNSL split is ours.** Jin *et al.* released no recognition split; our numbers define a new protocol, not replicate an existing one. We release the split JSON alongside the code.
2. **Cross-device / cross-signer is not solved.** Head anchoring cancels global translation and isotropic scale but not the combined nuisance of a new phone, a new signer's timing, and MediaPipe detection noise outside the training distribution. In a pilot cross-device probe (phone clips of a single sign from a held-out signer) none of our CNSL-5s models recovered the correct gloss in the top-5; this is a single-anecdote data point, not a benchmark result, but it flags that closing the in-the-wild gap will require more than geometric preprocessing.
3. **2D only.** We do not lift to 3D and cannot handle out-of-plane rotations directly. We argue that 2D-only avoids a known failure mode in ego imagery [16]; this is a contention, not a certainty.
4. **Shoulder scale is fragile under shoulder occlusion.** We currently guard against zero-division with a floor $\epsilon$=0.1 and otherwise leave the raw coordinates untouched for such frames. A principled fallback (e.g. torso-height estimate or running mean over detected frames) is left to future work.

# 8 Conclusion

HeaRing trades a learned disentangler for a two-line coordinate change. On the first published recognition baseline for the 5,794-class NationalCSL-DP benchmark, head anchoring alone recovers a **+20.68 pp** top-1 (**+41 % relative**) gain at zero extra parameters; on a 5-signer subset with a 12-layer hand-only backbone and robust augmentation we reach 69.93 % / 87.98 % top-1/5. The residual gap to in-the-wild phone footage — which remains large — points to future work in view-synthesis augmentation composed with head anchoring, not in replacing it.

# Acknowledgments

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